

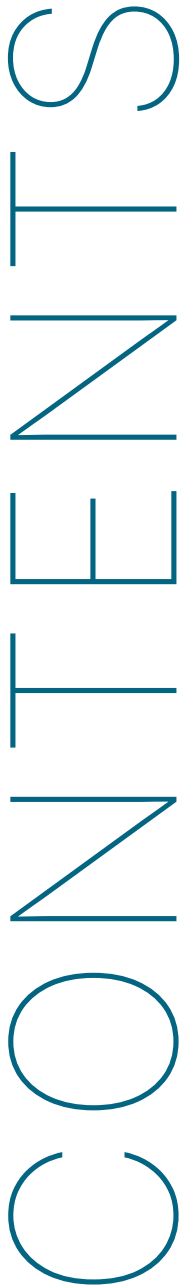
# Data Analysis Report

# **COHORT ANALYSIS**

PREPARED BY  
NHAN NGUYEN

PROJECT ON TRAINING  
MAY 2026

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**Cohort Analysis** is a behavioral analysis technique that divides users into groups based on shared characteristics over a specific period of time. It enables businesses to compare different groups of users, identify trends, observe behavioral patterns, and understand the underlying causes of customer actions.

This is an important analytical tool that helps businesses:

- Measure **Retention Rate**: What percentage of customers return after their first purchase?
- Understand **Product-Market Fit**: Can the product retain customers over time?
- Optimize the **Customer Journey**: At which stage do customers stop engaging with the business?
- Assess **Feature Impact**: Do new features or initiatives improve customer retention?

Three common types of Cohort:

- **Time-based Cohorts**: Groups based on time periods (e.g., registration month, first purchase month)
- **Behavior-based Cohorts**: Groups based on user actions (e.g., used feature X, referred friend)
- **Acquisition Channel Cohorts**: Groups based on acquisition sources (e.g., Google Ads, Organic, Referral)

In this project, the analysis focuses on **Time-based Cohorts**, where customers are grouped according to the month of their first purchase.

## Business Questions

This analysis aims to address the following business questions:

1. How effectively does the business retain customers after their first purchase?
2. At which point in the customer lifecycle does the largest drop-off occur?
3. Do newer customer cohorts show improvement in retention compared to earlier cohorts?
4. Is the current growth driven by sustainable customer behavior or repeated acquisition of new users?

## CHAPTER I

# OVERVIEW OF COHORT ANALYSIS

## 01 Introducing Cohort Analysis

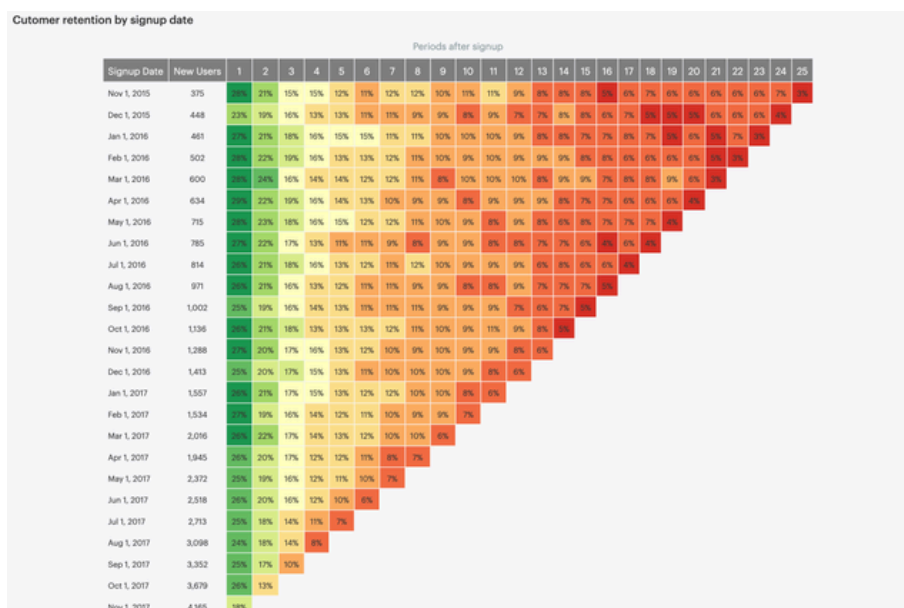
**Cohort Analysis** is a behavioral analytics technique that groups customers into cohorts based on a shared characteristic over a specific period of time. In customer retention analysis, cohorts are commonly formed according to the customer's first purchase date.

Unlike aggregate metrics, which summarize the behavior of all customers together, cohort analysis enables businesses to track and compare the behavior of different customer groups over time. This provides deeper insights into **customer retention, repeat purchase behavior, and long-term engagement.**

By analyzing customer cohorts, businesses can:

- Measure customer retention over time.
- Identify patterns of customer churn.
- Compare the performance of different customer cohorts.
- Evaluate the effectiveness of customer retention efforts.
- Better understand customer purchasing behavior.

Cohort analysis is widely used in e-commerce because it helps businesses understand whether customers continue purchasing after their first transaction. By tracking how customer activity changes over time, companies can identify opportunities to improve retention and encourage repeat purchases.

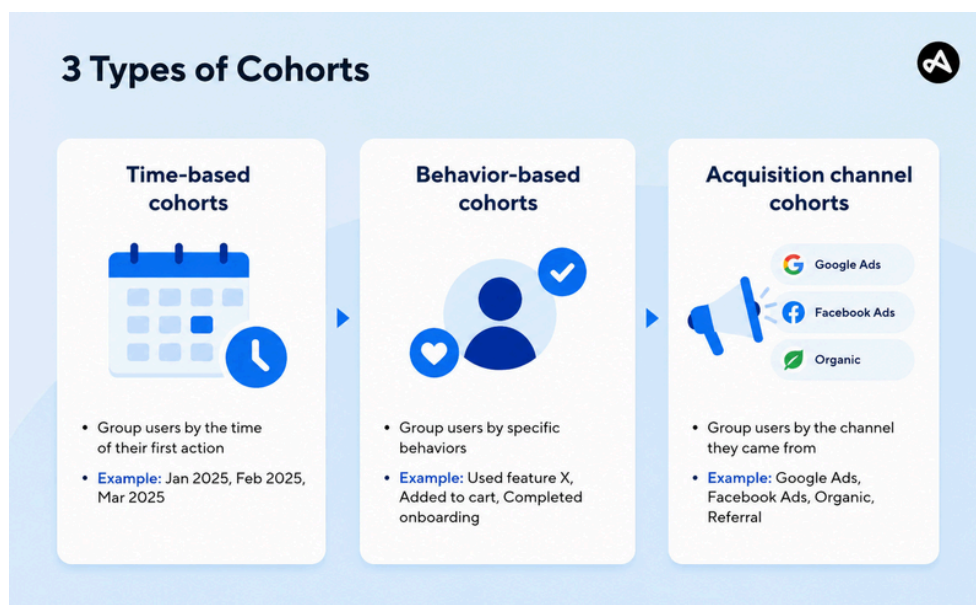


(Image source: Digital Gremlins)

## 02 Types of Cohort Analysis

Cohort Analysis is grouping framework, so how companies define cohorts determines the insights. There are three main types:

|                                    | Definition           | Measure                                                                                                                | Example                                                                                                                                                                                                                   |
|------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Time-based Cohorts</b>          | First action         | <ul style="list-style-type: none"> <li>Retention over time</li> <li>User quality across acquisition periods</li> </ul> | <ul style="list-style-type: none"> <li>Acquisition Cohort: Registration month (Jan 2026, Feb 2026)</li> <li>Purchase Cohort: Month of first purchase</li> <li>Activation Cohort: Week of onboarding completion</li> </ul> |
| <b>Behavior-based Cohorts</b>      | Specific action      | <ul style="list-style-type: none"> <li>Feature impact</li> <li>Product stickiness</li> </ul>                           | <ul style="list-style-type: none"> <li>Users who used feature A vs. those who did not</li> <li>Users who added to cart vs. those who did not</li> </ul>                                                                   |
| <b>Acquisition channel cohorts</b> | Acquisition channels | <ul style="list-style-type: none"> <li>Traffic quality</li> <li>Marketing ROI</li> </ul>                               | <ul style="list-style-type: none"> <li>Facebook Ads</li> <li>Google Ads</li> <li>Organic</li> <li>Referral</li> <li>Direct</li> </ul>                                                                                     |



## 03 Structure of Time-based Cohort Matrix

In this analysis, we focus on time-based cohorts, combined with a case study examining customer repeat purchase rates for an eCommerce store over time.

A cohort chart is a two-dimensional heatmap matrix where:

### ➤ Rows: Cohort

(Sample Cohort Analysis Structure)



| Signup Date | New Users | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|-------------|-----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Nov 1, 2015 | 375       | 28% | 21% | 15% | 15% | 12% | 11% | 12% | 12% | 10% | 11% | 11% | 9%  | 8% | 8% | 8% | 5% | 6% | 7% | 6% | 6% | 6% | 6% | 6% | 7% | 3% |
| Dec 1, 2015 | 448       | 23% | 19% | 16% | 13% | 13% | 11% | 11% | 9%  | 9%  | 8%  | 9%  | 7%  | 7% | 8% | 8% | 6% | 7% | 5% | 5% | 5% | 6% | 6% | 6% | 4% |    |
| Jan 1, 2016 | 461       | 27% | 21% | 18% | 16% | 15% | 15% | 11% | 11% | 10% | 10% | 10% | 9%  | 8% | 8% | 7% | 7% | 8% | 7% | 5% | 6% | 5% | 7% | 3% |    |    |
| Feb 1, 2016 | 502       | 28% | 22% | 19% | 16% | 13% | 13% | 12% | 11% | 10% | 9%  | 10% | 9%  | 9% | 9% | 8% | 8% | 6% | 6% | 6% | 6% | 5% | 3% |    |    |    |
| Mar 1, 2016 | 600       | 28% | 24% | 16% | 14% | 14% | 12% | 12% | 11% | 8%  | 10% | 10% | 10% | 8% | 9% | 9% | 7% | 8% | 8% | 9% | 6% | 3% |    |    |    |    |
| Apr 1, 2016 | 634       | 29% | 22% | 19% | 16% | 14% | 13% | 10% | 9%  | 9%  | 8%  | 9%  | 9%  | 9% | 8% | 7% | 7% | 6% | 6% | 6% | 4% |    |    |    |    |    |

- Each row represents a cohort defined by a specific time period.
- In this table, each row reflects the behavior of a group of users starting from their first month of signup (first action).

Examples:

- Nov 1, 2015: all users who signed up in January 2015
- Dec 1, 2015: a different cohort, completely independent from others

### ➤ Columns: Time offset

(Sample Cohort Analysis Structure)



| Signup Date | New Users | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|-------------|-----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Nov 1, 2015 | 375       | 28% | 21% | 15% | 15% | 12% | 11% | 12% | 12% | 10% | 11% | 11% | 9%  | 8% | 8% | 8% | 5% | 6% | 7% | 6% | 6% | 6% | 6% | 6% | 7% | 3% |
| Dec 1, 2015 | 448       | 23% | 19% | 16% | 13% | 13% | 11% | 11% | 9%  | 9%  | 8%  | 9%  | 7%  | 7% | 8% | 8% | 6% | 7% | 5% | 5% | 5% | 6% | 6% | 6% | 4% |    |
| Jan 1, 2016 | 461       | 27% | 21% | 18% | 16% | 15% | 15% | 11% | 11% | 10% | 10% | 10% | 9%  | 8% | 8% | 7% | 7% | 8% | 7% | 5% | 6% | 5% | 7% | 3% |    |    |
| Feb 1, 2016 | 502       | 28% | 22% | 19% | 16% | 13% | 13% | 12% | 11% | 10% | 9%  | 10% | 9%  | 9% | 9% | 8% | 8% | 6% | 6% | 6% | 6% | 5% | 3% |    |    |    |
| Mar 1, 2016 | 600       | 28% | 24% | 16% | 14% | 14% | 12% | 12% | 11% | 8%  | 10% | 10% | 10% | 8% | 9% | 9% | 7% | 8% | 8% | 9% | 6% | 3% |    |    |    |    |
| Apr 1, 2016 | 634       | 29% | 22% | 19% | 16% | 14% | 13% | 10% | 9%  | 9%  | 8%  | 9%  | 9%  | 9% | 8% | 7% | 7% | 6% | 6% | 6% | 4% |    |    |    |    |    |

- The first column (Signup Date) indicates when the cohort was created and serves as the baseline time point ( $t = 0$ ) for each row.
- The second column (New Users) shows the number of users in that cohort, helping assess its size.

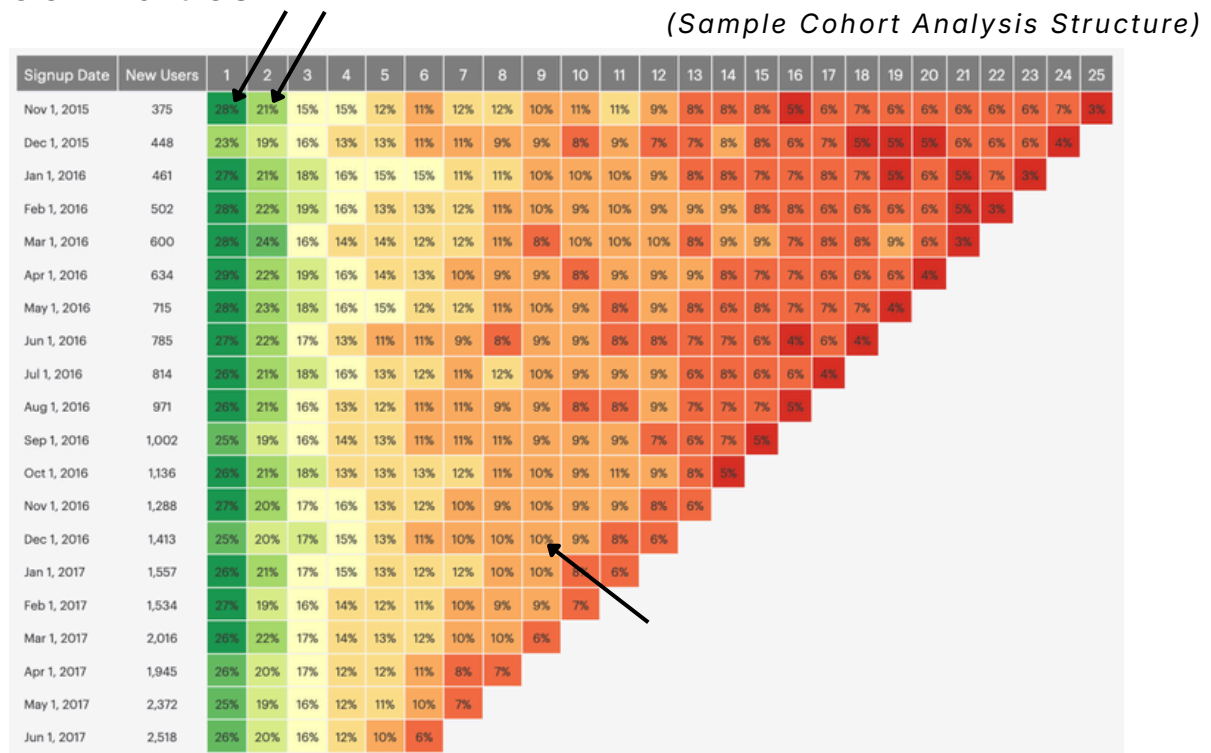
Examples:

- Nov 1, 2015: 375 new users
- Dec 1, 2015: 448 new users
- The subsequent columns (Periods after Signup: 1 → 25) represent the number of days elapsed since the baseline time.

Example:

- Column "1": Day 1 after signup

## ➤ Cell values



Each cell in the cohort chart contains tracked metrics. Common examples include:

- Retention Rate (%)
- Active Users
- Revenue per User

Example:

- The cohort on Nov 1, 2025 has a Day 1 retention rate of 28%, meaning 28% of the 375 users returned on Day 1.
- By Day 2, this decreases to 21% of the same 375 users.

We can generally observe that retention tends to decline over time, following a typical decay pattern.

## ➤ Structure

Most cohort charts have a triangular (diagonal) structure.

This is not due to missing data, but rather because newer cohorts have not existed long enough to be observed over later periods (e.g., Day 10, Day 20).

In addition, since this is a triangular heatmap, color is used to represent the magnitude of retention rates:

- Darker blue indicates higher retention
- Colors gradually fade toward red, indicating lower retention

## 04 Benefits Of Time-based Cohort Analysis

Time-based cohort analysis measures user quality across the lifecycle by isolating users based on when they started, allowing clear visibility into retention and changes over time.

**01 Remove time-mixing bias**  
Aggregate metrics such as overall retention mix users from different time periods, which leads to misleading conclusions, whereas cohort analysis isolates each user generation and enables accurate measurement of user quality over time.

**02 Accurately measure lifecycle retention**  
This approach clearly identifies where users drop off across the lifecycle, such as at Day 1 due to onboarding issues, at Day 7 due to a lack of habit formation, or in the long term due to insufficient core value, patterns that cannot be observed when only analyzing aggregate metrics like DAU or MAU.

**03 Compare user quality across acquisition periods**  
By comparing cohorts such as January, February, and March, this method allows you to determine whether newer users are better or worse and whether growth is accompanied by consistent or declining user quality.

**04 Detect the impact of product changes**  
When features are released or onboarding flows and pricing strategies are modified, comparing cohorts before and after these changes makes it possible to immediately observe shifts in retention and assess their impact.

**05 Identify seasonality effects**  
Users acquired during sales events, holidays, or major campaigns often exhibit different retention patterns, and cohort analysis helps account for these variations, preventing incorrect conclusions about overall product quality.



## 01 Data separation

## BRAZILIAN E-COMMERCE PUBLIC DATASET BY OLIST

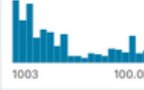
## Description of data sheets

The dataset has information of **100k orders from 2016 to 2018** made at multiple marketplaces in Brazil.

In this project, we have two tables that need to be analyzed:

 **Customers Table**

Provides customer-level information.

| customer_id                                                     | customer_unique_id               | customer_zip_code_prefix                                                           | customer_city                                           | customer_state                        |
|-----------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------|
| key to the orders dataset. Each order has a unique customer_id. | unique identifier of a customer. | first five digits of customer zip code                                             | customer city name                                      | customer state                        |
| 99441<br>unique values                                          | 96096<br>unique values           |  | sao paulo 16%<br>rio de janeiro 7%<br>Other (77019) 77% | SP 42%<br>RJ 13%<br>Other (44843) 45% |
| 06b8999e2fba1a1fbc88172c00ba8bc7                                | 861eff4711a542e4b93843c6dd7febb0 | 14489                                                                              | franca                                                  | SP                                    |
| 18955e83d337fd6b2def6b18a428a77                                 | 298c77bc529b7ac935b93aa66c333dc3 | 09790                                                                              | sao bernardo do campo                                   | SP                                    |
| 4e7b3e00285856ebd88712fd08374a83                                | 060e732b5b29e8181a18229c7b0b2b5e | 01151                                                                              | sao paulo                                               | SP                                    |
| b2b6027bc5c5109e529d4dc6358b12c3                                | 259dac757896d24d7702b9acbbf3f3c  | 08775                                                                              | mogi das cruzeiras                                      | SP                                    |
| 4f2d8ab171c80ec8364f7c12e35b23ad                                | 345ecd01c38d18a9036ed96c73b8d066 | 13056                                                                              | campinas                                                | SP                                    |
| 879864dab9bc3847522c92c82e1212b8                                | 4c93744516667ad3b8f1fb645a3116a4 | 89254                                                                              | jaraguá do sul                                          | SC                                    |

- **Grain:** One row represents a unique customer\_id
- **Key columns:**
  - customer\_id: primary key
  - customer\_unique\_id: identifies the actual user across multiple orders
- **Usage:** This table is used to identify unique customers and link them to their order history.

 **Orders Table**

Contains transactional data for each order placed on the platform.

| order_id                         | customer_id                                                       | order_status                                             | order_purchase_timestamp                                                            | order_approved_at                                                                    | order_delivered_at                                                                    | order_delivered_at                                                                    | order_estimated_delivery_at                                                             |
|----------------------------------|-------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| unique identifier of the order.  | key to the customer dataset. Each order has a unique customer_id. | Reference to the order status (delivered, shipped, etc). | Shows the purchase timestamp.                                                       | Shows the payment approval timestamp.                                                | Shows the order posting timestamp. When it was handed to the logistic partner.        | Shows the actual delivery date to the customer.                                       | Shows the estimated delivery date that was informed to customer at the purchase moment. |
| 99441<br>unique values           | 99441<br>unique values                                            | delivered 97%<br>shipped 1%<br>Other (1056) 2%           |  |  |  |  |    |
| a481f510dc54678b7cc4913af50a47   | 9ef432a06251297084e761801b0050d                                   | delivered                                                | 2017-10-02 10:56:33                                                                 | 2017-10-02 11:07:15                                                                  | 2017-10-04 19:55:00                                                                   | 2017-10-10 21:25:13                                                                   | 2017-10-10 00:00:00                                                                     |
| 53a0b2f0b0730c0b674142150279451  | 80820f747474dc6d20de480c80207ef                                   | delivered                                                | 2018-07-24 20:41:37                                                                 | 2018-07-26 03:24:27                                                                  | 2018-07-26 14:31:00                                                                   | 2018-08-07 15:27:45                                                                   | 2018-08-13 00:00:00                                                                     |
| 47779b09100c0804494d0cfd0cc5d    | 41cc3f5a00007f3443c3091a3a7009                                    | delivered                                                | 2018-08-08 00:30:49                                                                 | 2018-08-08 00:55:23                                                                  | 2018-08-08 13:50:00                                                                   | 2018-08-17 10:06:29                                                                   | 2018-09-04 00:00:00                                                                     |
| 949d5044d0f5de918fe9c16f97b45f8a | f80197465ea7920ad0bec7375364682                                   | delivered                                                | 2017-11-18 19:28:06                                                                 | 2017-11-18 19:45:59                                                                  | 2017-11-22 13:39:59                                                                   | 2017-12-02 00:28:42                                                                   | 2017-12-15 00:00:00                                                                     |
| ad21c59c08040dc03a9ceb5573f8159  | 8ab97984e4deaa886db0b47b7aad2c                                    | delivered                                                | 2018-02-13 21:18:39                                                                 | 2018-02-13 22:20:29                                                                  | 2018-02-14 19:46:34                                                                   | 2018-02-16 10:17:02                                                                   | 2018-02-26 00:00:00                                                                     |
| a4591c265e18cb1dc0e5209e20acc3   | 503740b9ca751cc0d67b207e9ab87600                                  | delivered                                                | 2017-07-09 21:57:05                                                                 | 2017-07-09 22:10:13                                                                  | 2017-07-11 14:50:04                                                                   | 2017-07-26 10:57:55                                                                   | 2017-08-01 00:00:00                                                                     |
| 13c0ce77aa42f0b0c0f53f0c79a0090  | e00271e0b7d0a00a93379590e7b727a                                   | invoiced                                                 | 2017-04-11 12:22:00                                                                 | 2017-04-13 13:25:17                                                                  |                                                                                       |                                                                                       | 2017-05-09 00:00:00                                                                     |

- **Grain:** One row represents a single order
- **Key columns:**
  - order\_id: unique identifier of the order
  - customer\_id: links to the customer table
  - order\_purchase\_timestamp: timestamp of purchase
- **Usage:** This table is used to track customer purchase behavior over time and serves as the primary source for cohort analysis.

## 02 Build Cohort Retention Framework in SQL

**Tool:** Google BigQuery - SQL

**How to do:**

- Process transactional order data from Olist dataset
- Identify each customer's first purchase month
- Calculate customer retention across monthly cohorts
- Use SQL functions: DATE\_TRUNC, DATE\_DIFF, WINDOW FUNCTIONS

### STEP 1 — Calculate Cohort Month

#### Cohort Month

The month when a customer made their first purchase.

#### Month Index

The number of months between the first purchase and subsequent purchases.

| Row | customer_unique_id            | COHORT_MONTH | month_index |
|-----|-------------------------------|--------------|-------------|
| 1   | 32ea3bdeab835c3aa6cb68ce...   | 2016-10-01   | 20          |
| 2   | ccafc1c3f270410521c3c6f3b2... | 2016-10-01   | 20          |
| 3   | d8f3c4f441a9b59a29f977df16... | 2017-01-01   | 19          |
| 4   | 87b3f231705783eb2217e2585...  | 2016-10-01   | 19          |
| 5   | 94e5ea5a8c1bf546db2739673...  | 2016-10-01   | 19          |
| 6   | 408aee96c75632a92e5079eee...  | 2017-01-01   | 17          |
| 7   | 5eefb861d4921a3e628bbc65c5... | 2017-03-01   | 17          |
| 8   | 4e23e1826902ec9f208e8cc613... | 2016-10-01   | 17          |
| 9   | a262442e3ab89611b44877c7a...  | 2017-02-01   | 17          |
| 10  | 97258e1c1f77f32358eccd1c9e... | 2017-03-01   | 17          |
| 11  | 18bc87094128bbfe943cf88adc... | 2017-01-01   | 17          |
| 12  | 8f6ce2295bdec03cd50e34b4b...  | 2017-02-01   | 17          |
| 13  | 7e7301841ddb4064c2f3a31e4...  | 2017-02-01   | 17          |
| 14  | 1b6e96ed99cb8d135efe220d7...  | 2017-03-01   | 17          |
| 15  | 24072811917876a84c81166f9...  | 2017-01-01   | 17          |

### STEP 2 — Build Retention Matrix in SQL

We calculate customer retention by measuring the percentage of customers who return and make purchases in subsequent months after their first transaction.

#### FORMULA

$$\text{Retention Rate} = \frac{\text{Returning Customers}}{\text{Total Customers in Cohort}} \times 100\%$$

(In this project)

$$\text{Retention Rate} = \frac{\text{Users}}{\text{Cohort Size}} \times 100\%$$

| COHORT_MONTH | month_index | USERS | COHORT_SIZE | RETENTION_RATE    |
|--------------|-------------|-------|-------------|-------------------|
| 2016-09-01   | 0           | 4     | 4           | 1.0               |
| 2016-10-01   | 0           | 321   | 321         | 1.0               |
| 2016-10-01   | 6           | 1     | 321         | 0.003115264797... |
| 2016-10-01   | 9           | 1     | 321         | 0.003115264797... |
| 2016-10-01   | 11          | 1     | 321         | 0.003115264797... |
| 2016-10-01   | 13          | 1     | 321         | 0.003115264797... |
| 2016-10-01   | 15          | 1     | 321         | 0.003115264797... |
| 2016-10-01   | 17          | 1     | 321         | 0.003115264797... |
| 2016-10-01   | 19          | 2     | 321         | 0.006230529595... |
| 2016-10-01   | 20          | 2     | 321         | 0.006230529595... |
| 2016-12-01   | 0           | 1     | 1           | 1.0               |
| 2016-12-01   | 1           | 1     | 1           | 1.0               |
| 2017-01-01   | 0           | 764   | 764         | 1.0               |
| 2017-01-01   | 1           | 3     | 764         | 0.003926701570... |
| 2017-01-01   | 2           | 2     | 764         | 0.002617801047... |

# CHAPTER III

## CUSTOMER RETENTION ANALYSIS

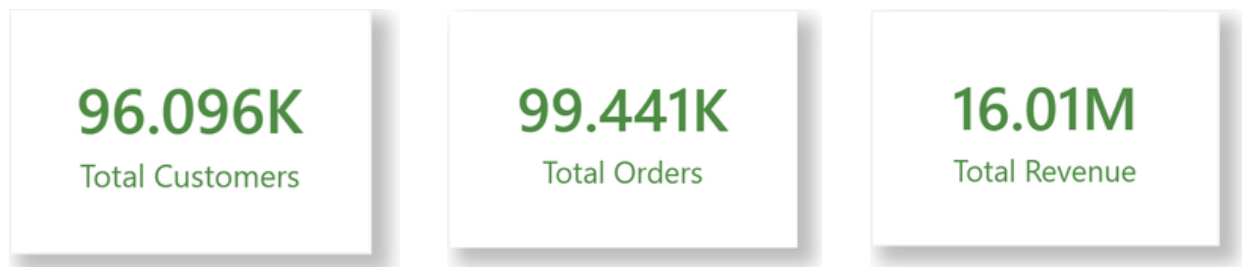
### 01 KPI Overview

Using Power BI 

#### »» Total Customers & Total Orders

Currently, the database records indicate that more than 96,000 customers have made purchases on the platform, generating a total of over 99,000 orders. In addition, the total revenue from customer transactions has exceeded 16 million BRL.

These figures demonstrate that the platform has successfully attracted a large customer base and maintained a considerable level of transaction activity throughout the observed period.



The relatively close gap between total customers and total orders suggests that a substantial proportion of customers may have **made only a limited number of purchases**.

This potentially indicates weak repeat purchase behavior and highlights the importance of analyzing customer retention patterns through cohort analysis.

#### »» Repeat Purchase Rate

The Repeat Purchase Rate measures the percentage of customers who returned to make more than one purchase on the platform.

##### FORMULA

$$\text{Repeat Purchase Rate} = \frac{\text{Customers With More Than One Purchase}}{\text{Total Customers}} \times 100\%$$

Currently, the platform records a Repeat Purchase Rate of only 3.1%, indicating that the **majority of customers made only a single purchase** during the observed period.

**3.12%**  
Repeat Purchase Rate

This suggests that while the platform demonstrates relatively strong customer acquisition performance, **long-term customer retention and repeat engagement remain limited.**

## »» Average Orders per Customer

The Average Orders per Customer metric measures the average number of orders generated by each customer on the platform.

### FORMULA

$$\text{Average Orders per Customer} = \frac{\text{Total Orders}}{\text{Total Customers}} \times 100\%$$

Currently, the platform records an average of only 1.03 orders per customer, indicating that most customers placed just a single order during the observed period.

**1.03**  
Average Orders Per Customer

This reflects relatively **weak repeat purchasing behavior** and suggests that long-term customer engagement remains limited across the platform.

## »» Average Retention Rate

The Average Month 1 Retention Rate measures the percentage of customers who returned to make another purchase one month after their initial transaction.

### FORMULA

$$\text{Month 1 Retention Rate} = \frac{\text{Customer Returning in Month 1}}{\text{Original Cohort Size}} \times 100\%$$

On average, only 5.22% of customers returned to make a second purchase in the month following their first purchase.

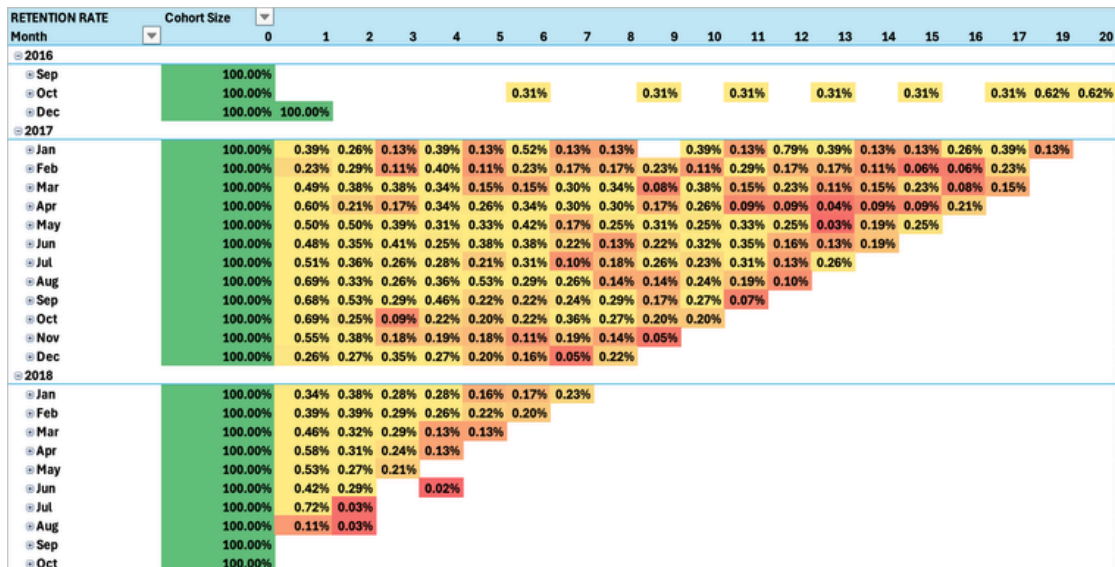
**5.22%**  
Average Retention Rate

In other words, approximately **95 out of every 100 customers did not return** within the first month after acquisition.

This indicates that the business experienced a significant drop in customer activity immediately after the first purchase, suggesting that **converting first-time buyers into repeat customers was a major challenge.**

## 02 Cohort Heatmap Overview

The cohort retention heatmap illustrates customer retention performance over time by tracking how many customers return in subsequent months after their first purchase.



(Monthly Customer Retention Cohort Analysis)



### Strongest & Weakest Cohorts

The **May to Sep-2017** cohorts show relatively higher customer retention compared to most other cohorts. Customers acquired during these periods were **more likely to return** and make additional purchases.

Meanwhile, several **early cohorts in 2017** experienced faster retention decline and lower returning customer counts.



### Stronger Mid-to-Late 2017 Cohorts

Some cohorts in mid-2017 show slightly better retention compared to other periods, but the overall retention trend remains relatively low.

This suggests that while the platform was able to attract many customers, **maintaining long-term customer engagement was more challenging**.



### Growing Customer Acquisition but Weak Retention

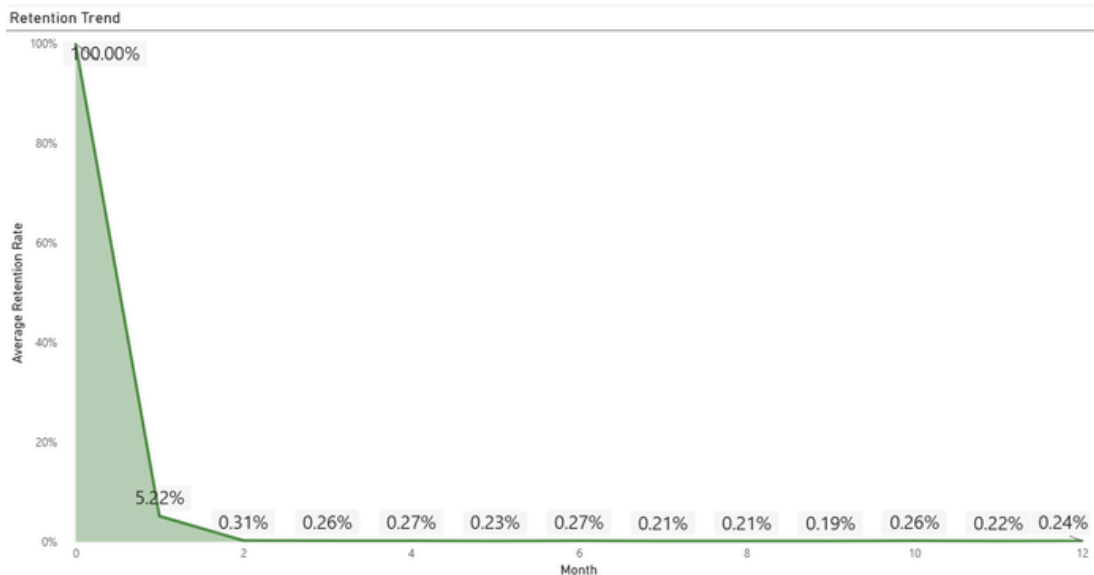
However, the number of customers acquired in Month 0 gradually **increased over time**.

For example, the cohort size increased from only 321 customers in October 2016 to more than 7,000 customers in early 2018 cohorts.

This indicates that the platform continued **attracting new customers consistently** during the observed period and experienced noticeable growth in customer acquisition activity.

| TOTAL USERS | Cohort Size |
|-------------|-------------|
| Month       | 0           |
| 2016        |             |
| Sep         | 4           |
| Oct         | 321         |
| Dec         | 1           |
| 2018        |             |
| Jan         | 7025        |
| Feb         | 6451        |
| Mar         | 6965        |

## 03 Retention Trend Analysis



### ➤ Sharp retention decline

The retention rate dropped sharply after the first purchase. It decreased from **100% in Month 0** to only **5.22% in Month 1**.

This means that most customers **did not return** to make another purchase after their first order.

### ➤ Early Customer Churn

Customer retention continued to decline during the first few months. The retention rate fell from 5.22% in Month 1 to around 0.31% in Month 3.

This shows that many customers **stopped purchasing shortly** after their first transaction, making the early stage of the customer lifecycle the most critical period for retention.

### ➤ Long-Term Customer Retention

After Month 4, the retention rate remained **relatively stable at around 0.25%**.

Although only a small number of customers continued purchasing over time, these customers remained active for a longer period and contributed to long-term business performance.

Overall, **long-term customer retention existed** but represented only a **small portion** of the customer base.

The platform appears to **perform well in customer acquisition** but **struggles to maintain long-term customer retention** and engagement.



## 04 Repeat Purchase Behavior

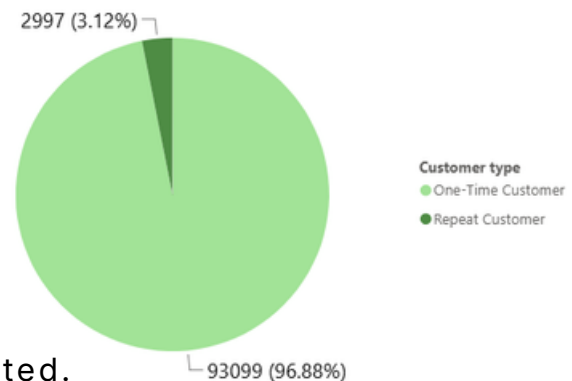
This section examines customer repurchase patterns to better understand customer engagement and loyalty on the platform.

By analyzing purchase frequency, we can evaluate how often customers return after their first transaction and identify opportunities to improve long-term customer retention.

### ➤ One-Time vs Repeat Customers

The majority of customers made only one purchase during the observed period.

Repeat customers account for a relatively small portion of the customer base.

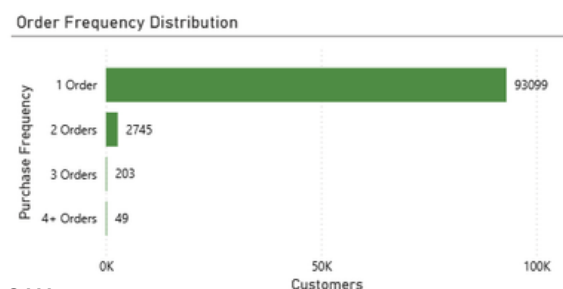


- Repeat purchasing behavior is limited.
- Customer loyalty remains relatively low.
- Acquiring customers appears easier than retaining them.

### ➤ Order Frequency Distribution

Customer counts decrease significantly as the number of purchases increases.

Only a small percentage of customers place multiple orders.



- Purchasing frequency is generally low.
- Highly engaged customers represent only a small segment of the customer base.
- Opportunities exist to encourage more frequent purchases from existing customers.

## SUMMARY

While the platform successfully attracted a large number of customers, only a small proportion returned to make additional purchases.

This pattern is consistent with the findings from the cohort retention analysis, which showed **low retention rates** across most cohorts and a **significant decline in customer activity after the first purchase**.



## CHAPTER IV

# BUSINESS INSIGHTS & RECOMMENDATIONS



**How effectively does the platform retain newly acquired customers after their first purchase?**



The cohort retention analysis shows that retention declines sharply after Month 0 across almost all cohorts. Most customers do not return after their initial purchase.



Strengthen post-purchase engagement through follow-up emails, personalized product recommendations, and customer feedback campaigns.



**Are customers making repeat purchases, or are they primarily one-time buyers?**



The repeat purchase analysis indicates that most customers placed only one order. The Average Orders per Customer is approximately 1.03, suggesting limited repeat purchasing behavior.



Implement loyalty programs and customer rewards to encourage repeat purchases and increase purchasing frequency.



**Is customer acquisition growth translating into long-term customer engagement?**

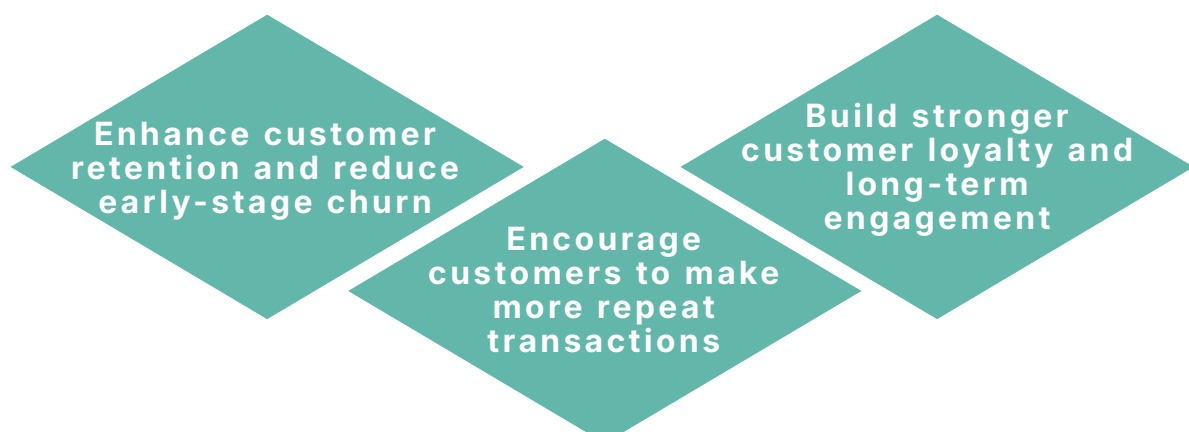


Although cohort sizes increased significantly over time, retention rates remained relatively low. This suggests that customer acquisition growth was not accompanied by strong customer retention.



Launch remarketing campaigns targeting inactive customers who have not returned after their first purchase.

## EXPECTED BUSINESS IMPACT





# APPENDIX

## A. SQL QUERIES

-- Build Cohort Retention Framework

```
WITH
BASE AS(
  SELECT
    ORD.order_id,
    C.customer_id,
    C.customer_unique_id,
    DATE(order_purchase_timestamp) AS ORDER_DATE
  FROM `Ecommerce.Orders` ORD
  LEFT JOIN `Ecommerce.Customers` C
  ON ORD.customer_id = C.customer_id
),
First_Purchase AS(
  SELECT
    C.customer_unique_id,
    MIN(DATE(order_purchase_timestamp)) AS FIRST_PURCHASE
  FROM `Ecommerce.Orders` ORD
  LEFT JOIN `Ecommerce.Customers` C
  ON ORD.customer_id = C.customer_id
  GROUP BY customer_unique_id),
Cohort_base AS (
  SELECT
    FP.customer_unique_id,
    B.order_id,
    FP.FIRST_PURCHASE,
    B.ORDER_DATE,
    DATE_DIFF(B.ORDER_DATE, FP.FIRST_PURCHASE, MONTH) AS month_index,
    DATE_TRUNC(FP.FIRST_PURCHASE, MONTH) AS COHORT_MONTH
  FROM First_Purchase FP
  LEFT JOIN Base B
  ON FP.customer_unique_id = B.customer_unique_id),
Retention_table AS(
  SELECT
    CB.month_index,
    CB.COHORT_MONTH,
    COUNT (DISTINCT CB.Customer_unique_id) AS USERS
  FROM Cohort_base CB
  GROUP BY CB.COHORT_MONTH, CB.month_index
),
Final AS(
  SELECT
    Retention_table.COHORT_MONTH,
    Retention_table.month_index,
    Retention_table.USERS,
    FIRST_VALUE(USERS) OVER (
      PARTITION BY Retention_table.COHORT_MONTH
      ORDER BY Retention_table.month_index
    ) AS COHORT_SIZE,
    Retention_table.USERS * 1.0 / FIRST_VALUE(USERS) OVER(
      PARTITION BY Retention_table.COHORT_MONTH
      ORDER BY Retention_table.month_index
    ) AS RETENTION_RATE
  FROM Retention_table
)
SELECT *
FROM Final
ORDER BY Final.COHORT_MONTH, Final.month_index
```

```

-- Repurchase Rate Calculation
WITH orders as
(SELECT
    C.customer_unique_id,
    count(order_id) as order_num
FROM `Ecommerce.Orders` O
LEFT JOIN `Ecommerce.Customers` C
ON O.customer_id = C.customer_id
group by (customer_unique_id))
SELECT
    COUNT
    (CASE
        WHEN ORDERS.order_num > 1
        THEN 1 END)*100.0/COUNT(*) AS Repurchase_Rate
FROM ORDERS

```

```

-- Average Orders Per Customer
SELECT
    COUNT (DISTINCT O.ORDER_ID)*1.0/
    COUNT(DISTINCT C.CUSTOMER_UNIQUE_ID) AS AOPC
FROM `Ecommerce.Orders` O
LEFT JOIN `Ecommerce.Customers` C
ON O.CUSTOMER_ID = C.CUSTOMER_ID

```

```

-- One Time vs. Repeat Customers
WITH customer_orders AS (
SELECT
    c.customer_unique_id,
    COUNT(DISTINCT o.order_id) AS order_count
FROM Ecommerce.Customers c
LEFT JOIN Ecommerce.Orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_unique_id
)

SELECT
CASE
    WHEN order_count = 1 THEN 'One-Time Customer'
    ELSE 'Repeat Customer'
END AS customer_type,
COUNT(*) AS customers
FROM customer_orders
GROUP BY customer_type

```

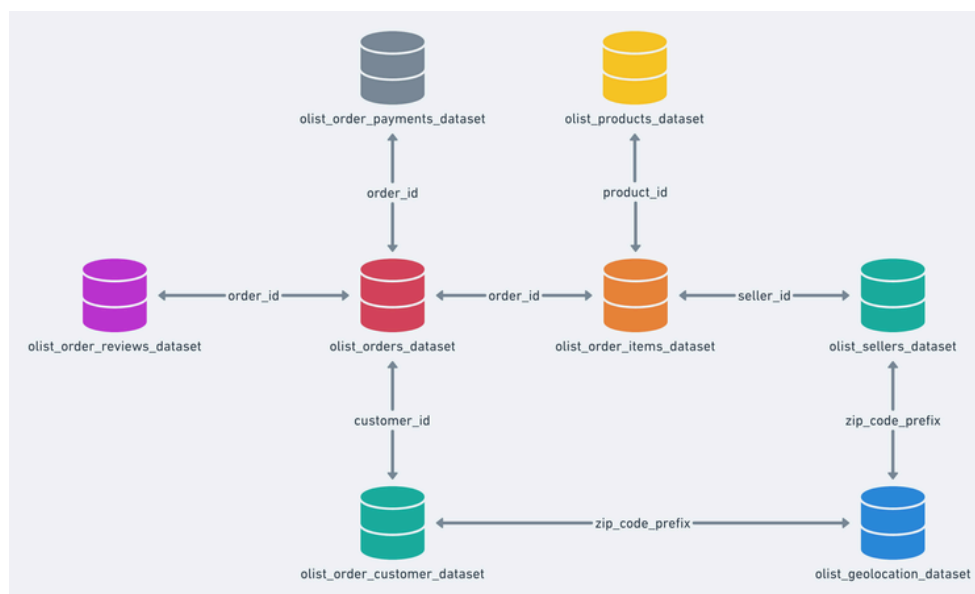
```

-- Order Frequency Distribution
WITH customer_orders AS (
SELECT
    c.customer_unique_id,
    COUNT(DISTINCT o.order_id) AS order_count
FROM Ecommerce.Customers c
LEFT JOIN Ecommerce.Orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_unique_id
)

SELECT
CASE
    WHEN order_count = 1 THEN '1 Order'
    WHEN order_count = 2 THEN '2 Orders'
    WHEN order_count = 3 THEN '3 Orders'
    ELSE '4+ Orders'
END AS purchase_frequency,
COUNT(*) AS customers
FROM customer_orders
GROUP BY purchase_frequency
ORDER BY purchase_frequency

```

## B. DATA MODEL



*Brazilian E-commerce by Olist dataset*

This project primarily uses the Customers, Orders, and Payments tables to construct the cohort retention framework and analyze customer purchasing behavior.

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**Nhan Nguyen**  
*Data Analyst*

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**Contact**

0916 996 296 

nhan.ngtg@gmail.com 